EXPERIMENT 4

Demonstrate data visualization, histograms and multiple variable summaries.

AIM

To demonstrate data visualization techniques using histograms, scatter plots, and multiple variable summaries on COVID-19 and other datasets.

PROCEDURE

1. Import pandas, matplotlib, sklearn (MinMaxScaler).
2. Load the worldometer_coronavirus_daily_data.csv.
3. Inspect dataset using data.tail() and data.info().
4. Drop nulls and duplicates; convert 'date' to datetime.
5. Select numeric columns: daily_new_cases, daily_new_deaths, active_cases, cumulative_total_cases.
6. Apply MinMaxScaler to normalize the numeric data.
7. Plot a scatter plot of Scaled Daily Cases vs Scaled Daily Deaths.
8. Plot a histogram of scaled daily new cases (xlim 0 to 0.1).
9. Compute descriptive statistics using scaled_data.describe().
10. Plot multiple variable box plots using scaled_data.plot(kind='box').
11. Compute and print the correlation matrix.
12. Visualize the correlation matrix using imshow with coolwarm colormap.

CODE

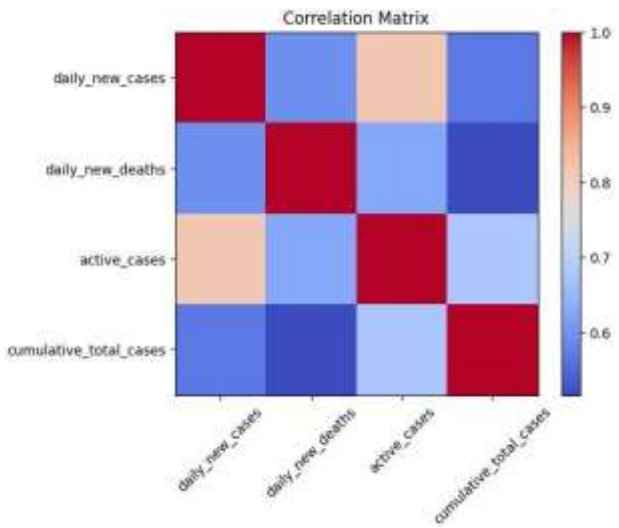
```
import pandas as pd import matplotlib.pyplot
as plt from sklearn.preprocessing import
MinMaxScaler

data = pd.read_csv('worldometer_coronavirus_daily_data.csv')
data.tail() data.info()

data = data.dropna() data =
data.drop_duplicates() data['date'] =
pd.to_datetime(data['date'])

numeric_data = data[['daily_new_cases','daily_new_deaths',
                      'active_cases','cumulative_total_cases']]

scaler = MinMaxScaler() scaled_data =
pd.DataFrame(
scaler.fit_transform(numeric_data),
columns=numeric_data.columns
)

# Scatter plot
plt.scatter(scaled_data['daily_new_cases'], scaled_data['daily_new_deaths'])
plt.xlabel('Scaled Daily Cases') plt.ylabel('Scaled Daily Deaths')
plt.title('Scatter Plot after Scaling') plt.show()

#                                         Histogram
plt.hist(scaled_data['daily_new_cases'], bins=30)
plt.xlim(0, 0.1)
```

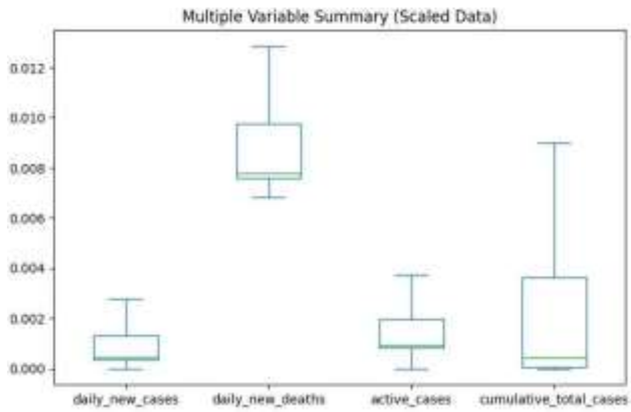
```
plt.xlabel('Scaled Daily Cases') plt.ylabel('Frequency')
plt.title('Histogram of Scaled Daily Cases') plt.show()

# Summary statistics
scaled_data.describe()

# Multiple variable boxplot scaled_data.plot(kind='box',
figsize=(8,5), showfliers=False) plt.title('Multiple Variable
Summary (Scaled Data)') plt.show()

# Correlation matrix
corr_matrix = numeric_data.corr() plt.imshow(corr_matrix, cmap='coolwarm')
plt.colorbar() plt.xticks(range(len(corr_matrix.columns)),
corr_matrix.columns, rotation=45) plt.yticks(range(len(corr_matrix.columns)),
corr_matrix.columns) plt.title('Correlation Matrix') plt.show()
```

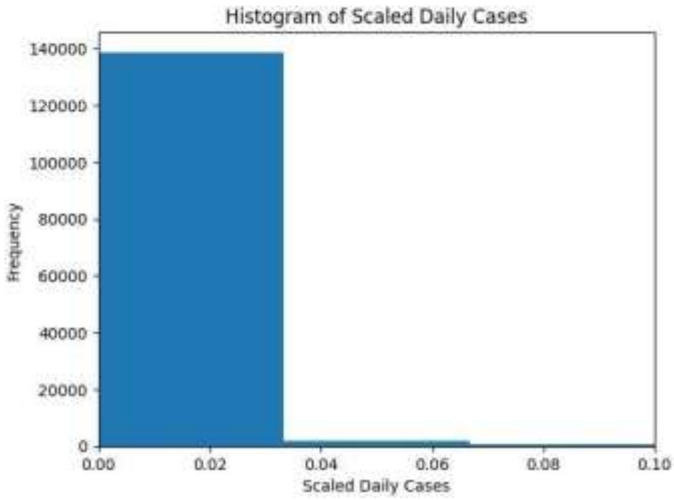
OUTPUT Scaled Data Summary (describe): count: 142034 rows for all columns mean daily_new_cases: 0.004067

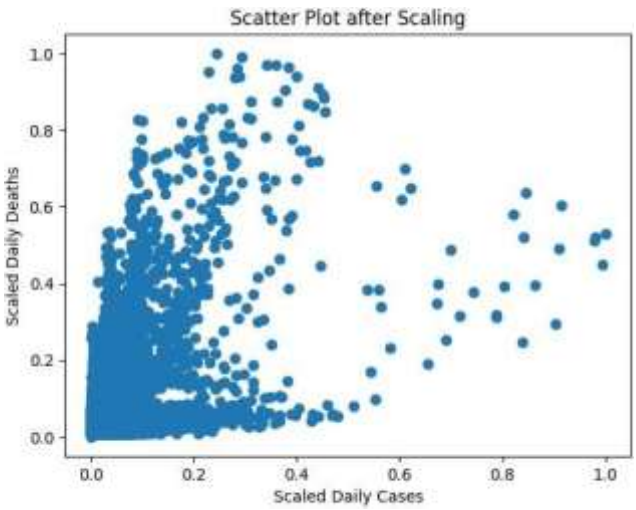
mean daily_new_deaths: 0.015478

Correlation Matrix highlights:
daily_new_cases vs active_cases: 0.810888
daily_new_cases vs cumulative_total_cases: 0.568780









RESULT

Data visualization revealed a right-skewed distribution in daily new cases. The correlation matrix showed a strong positive correlation (0.81) between daily new cases and active cases. MinMax scaling normalized the data for fair cross-variable comparison.